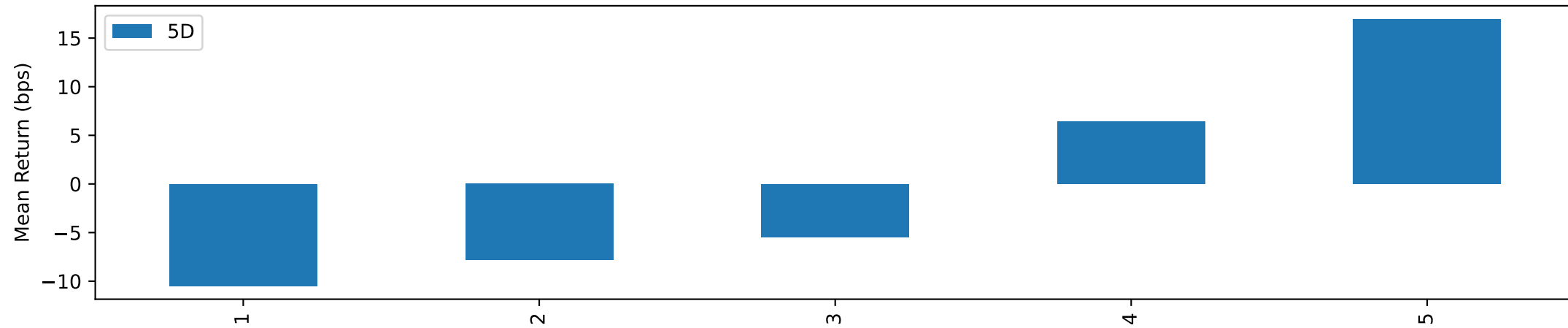
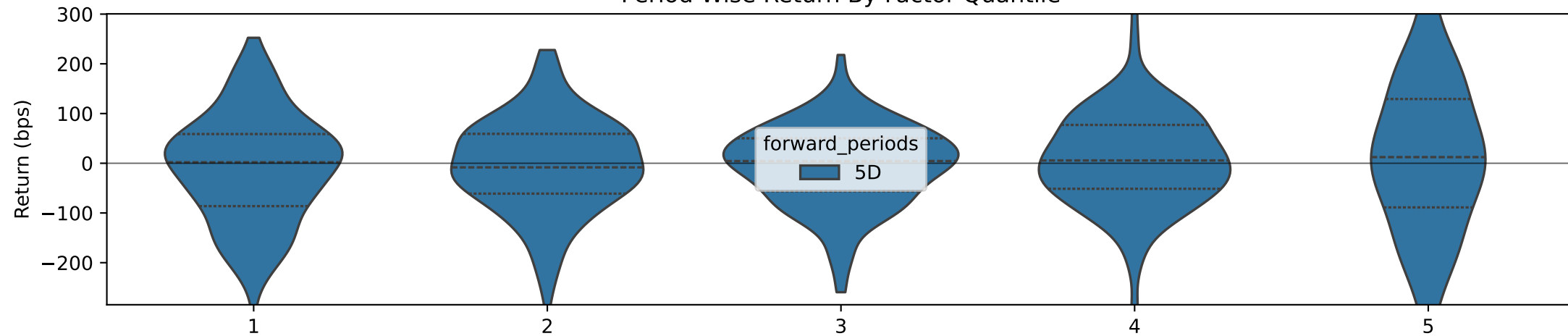


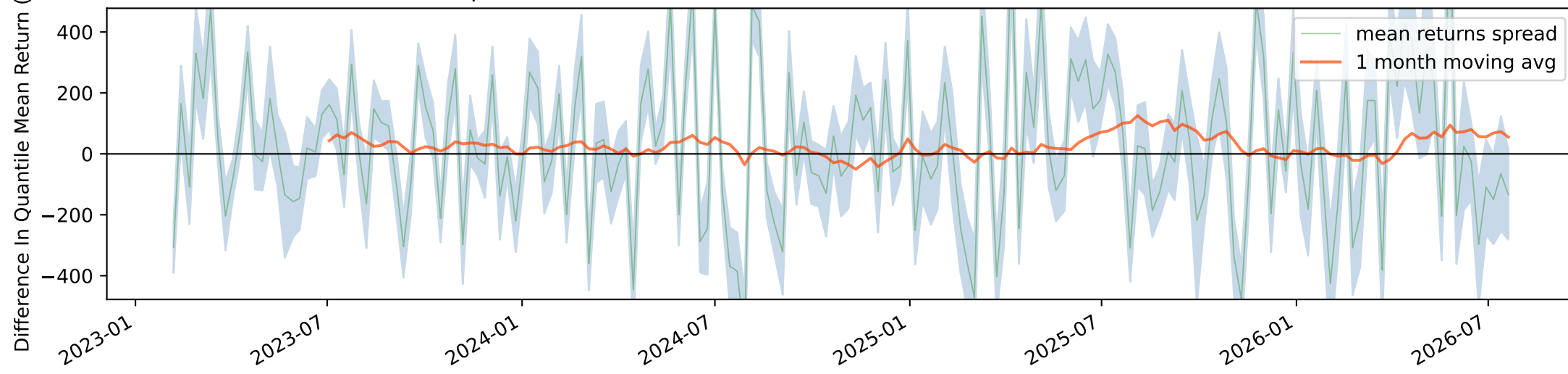
Mean Period Wise Return By Factor Quantile



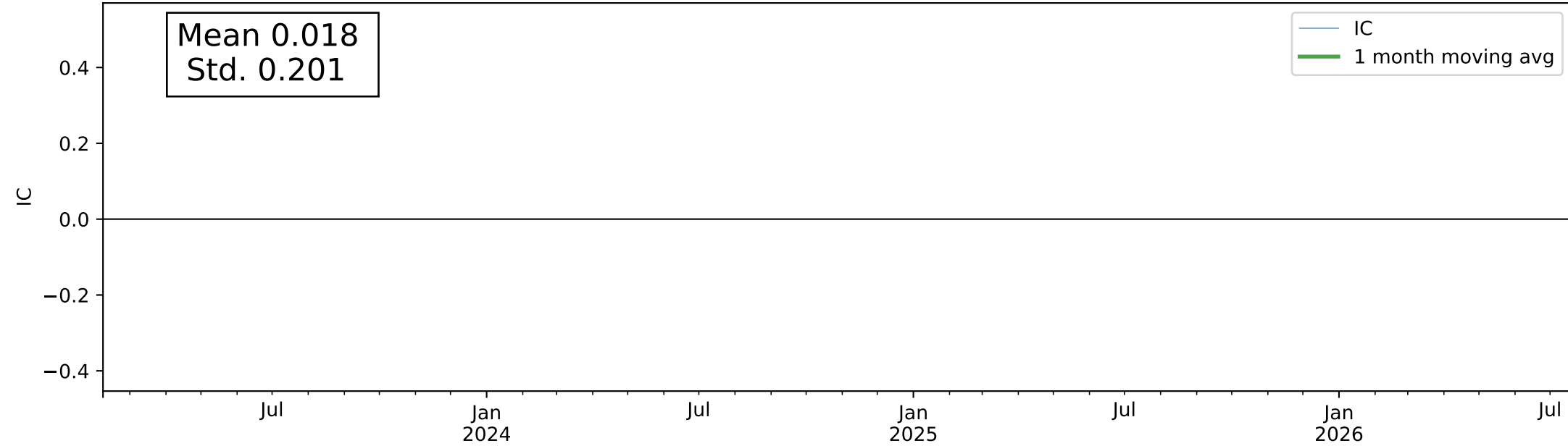
Period Wise Return By Factor Quantile



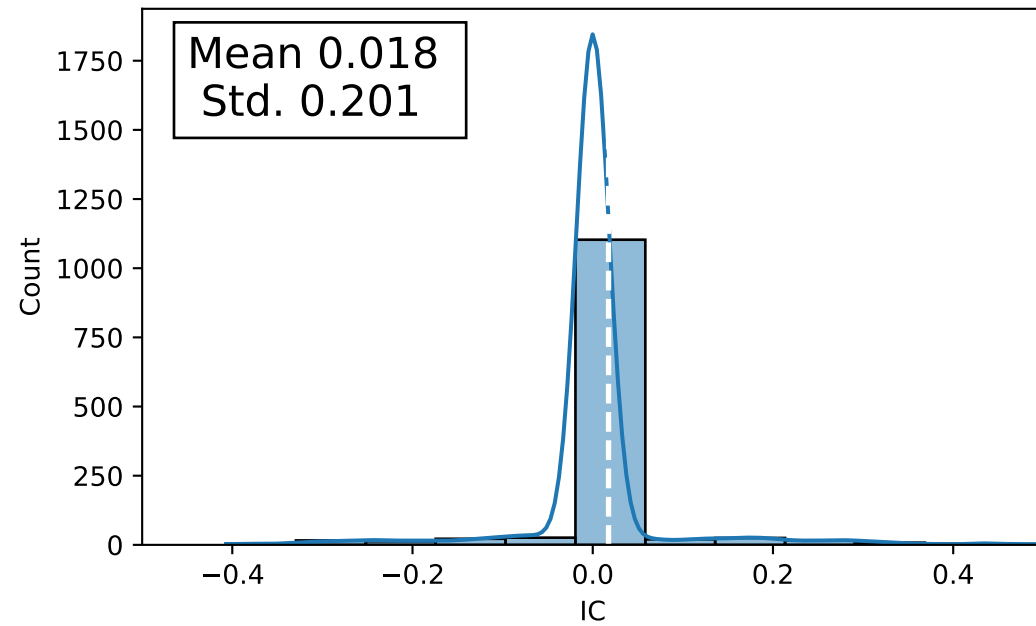
Top Minus Bottom Quantile Mean Return (5D Period Forward Return)



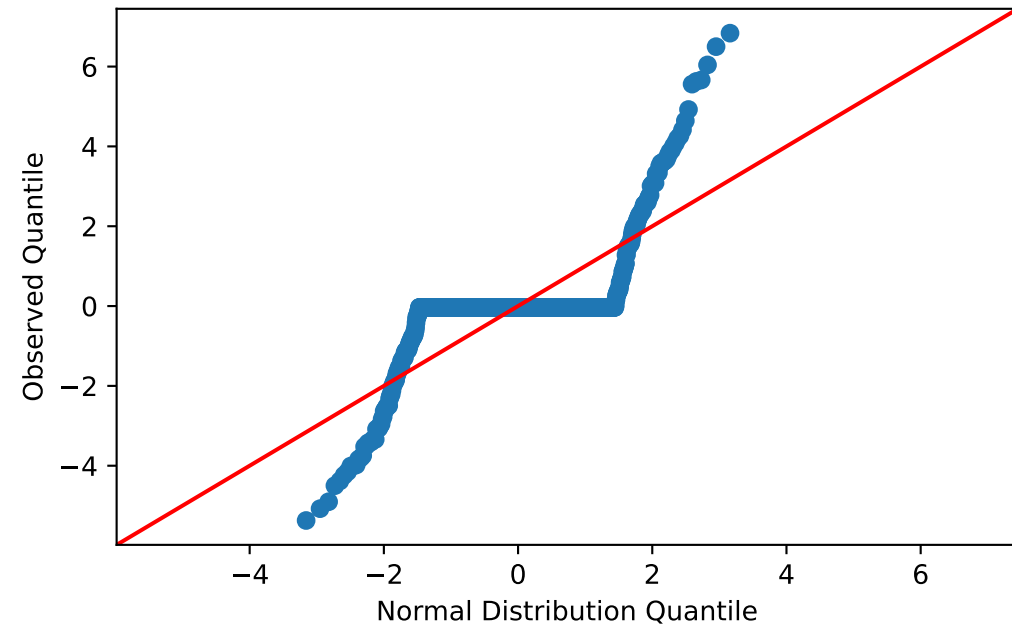
5D Period Forward Return Information Coefficient (IC)



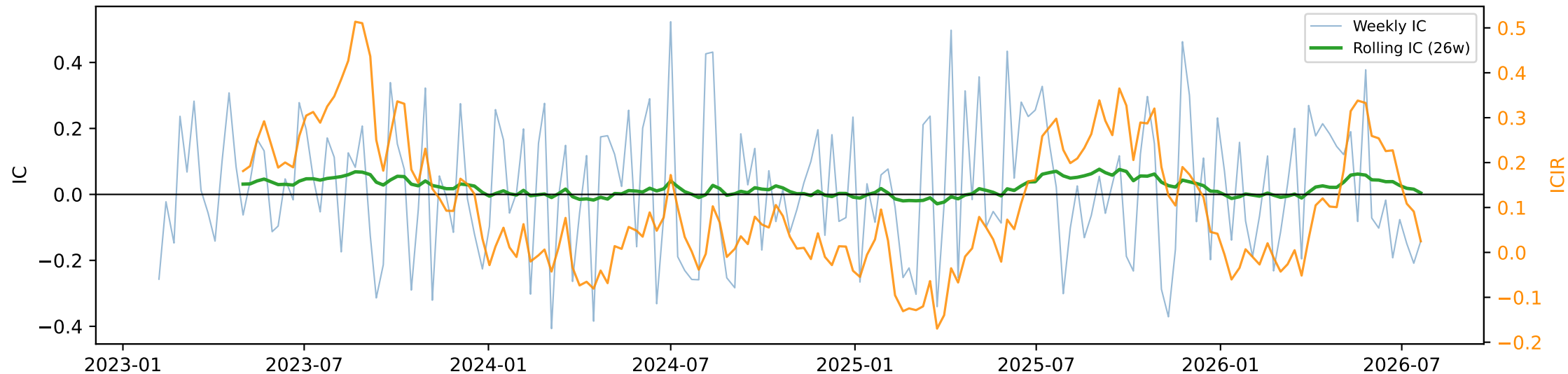
5D Period IC



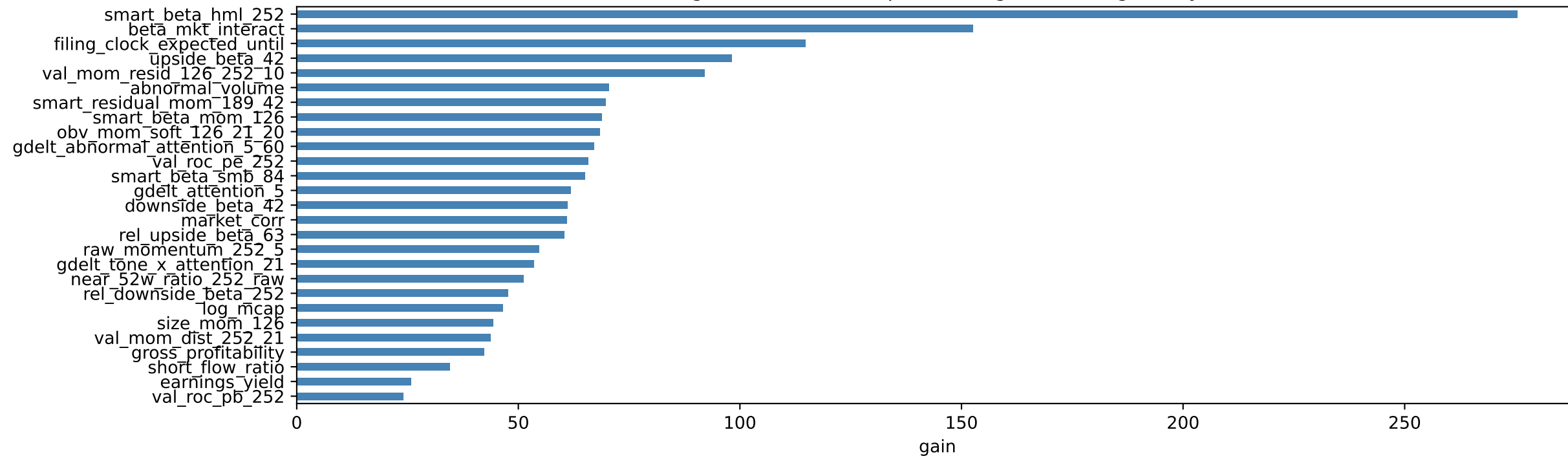
5D Period IC Normal Dist. Q-Q



Holdout weekly IC and rolling mean IC

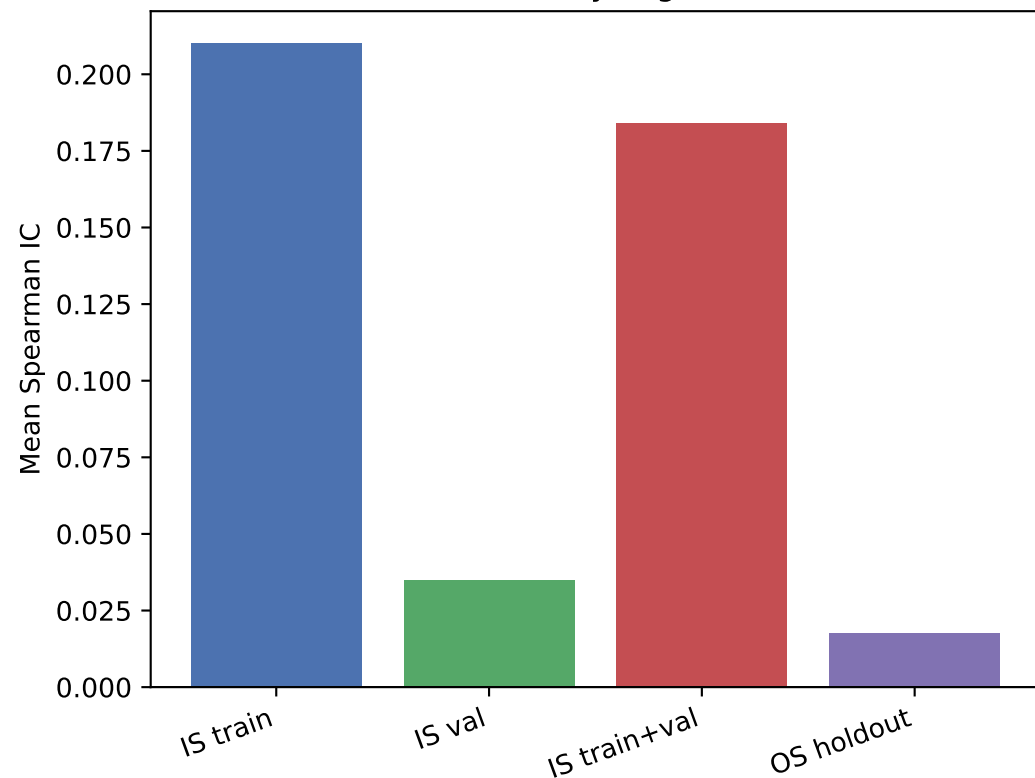


LightGBM feature importance (gain) — insight only

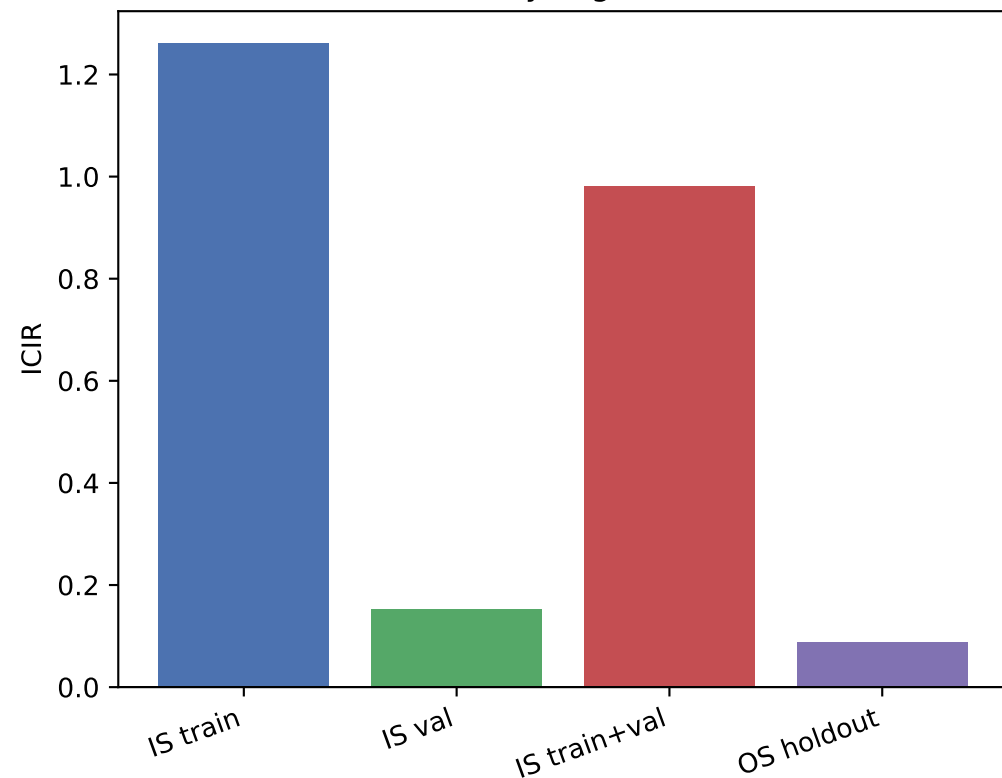


S1 LightGBM Tearsheet | IS IC=0.1838/0.98 OS IC=0.0175/0.09 trees=67

Mean IC by segment



ICIR by segment



IS train+val: IC=0.1838, ICIR=0.981 (n=421) | OS holdout: IC=0.0175, ICIR=0.087 (n=181)

S1 LightGBM Tearsheet | IS IC=0.1838/0.98 OS IC=0.0175/0.09 trees=67

Model: s1_lgbm_is.txt (IS-only scores)

IC / ICIR comparison (Spearman vs fwd_ret_5):
IS train: IC=0.2101 ICIR=1.261 n=358
IS val: IC=0.0348 ICIR=0.153 n=63
IS train+val: IC=0.1838 ICIR=0.981 n=421
OS holdout: IC=0.0175 ICIR=0.087 n=181

Best iteration: 67
BEST_PARAMS: {'max_depth': 6, 'num_leaves': 31, 'learning_rate': 0.01, 'min_child_samples': 350, 'subsample': 0.8546741472798252, 'colsample_bytree': 0.9548850970305306, 'reg_lambda': 0.1}